

Project 1b

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ABSTRACT

[Summarize the named audience, verified market gap, proposed product, one-month scope, measurable benefit, and LLM-assisted evidence process.]

1 INTRODUCTION

WolfBite is a Flutter/Firebase shopping assistant for WIC participants. In Project 1a, the team reverse engineered its behavior and independently tested its principal user flows. This report uses market evidence and independent LLM analysis to decide whether to extend WolfBite or pivot, then defines a product that four students can build and test in one month.

[State the final decision and preview the report's contribution.]

2 D1: MARKET SURVEY

2.1 Method

We ran the shared market-survey Prompt 1 with four LLM analysts: Codex, Terra, Gemini, and local Ollama, to generate rival candidates. We retained a rival only when at least two models named it or one model supplied a live, relevant URL. We then checked official product sources and independently audited the three strongest rivals. Each final rival independently satisfies the live-URL qualification rule through an official product source.

2.2 Verified Rivals

Table 1 summarizes the three verified rivals.

2.3 Selected Gap and Audience

Gap: Our feature audit found no documented integrated, item-level explanation-and-recovery workflow in WICShopper, myWIC Mosaic, or Bnft for cases where an app indicates that a product is eligible but checkout rejects it. Such a workflow could connect the rejected item, the participant's current benefit or food category, and purchase evidence to likely causes and actionable next steps.

Audience: WIC caregivers and participants shopping for their households, especially those managing young children while an item they expected to be covered is rejected at checkout.

This need is supported by participant research reporting difficulty identifying WIC-eligible foods and stigma during shopping and checkout [1, 2]. WIC caregivers also rank balance checking and barcode scanning among their highest-priority app features

[7]. Together, this evidence suggests that eligibility uncertainty during shopping remains consequential for the target audience.

We also evaluated two alternative gaps. We rejected nutrition comparison because participant evidence for that need was comparatively weak. We rejected broader receipt reconciliation because all three rivals already provide some form of purchase or transaction history. The narrower explanation-and-recovery gap therefore has stronger support from both the competitor audit and participant evidence.

2.4 Counterevidence and Scope

The conclusion has **medium confidence**. WICShopper already provides item-level purchase history in some agencies, myWIC tracks purchases, and Bnft provides transaction history. Public documentation may also omit program-specific features. In one month, the team can build and test a WolfBite prototype using fixed mismatch scenarios, rule-based likely causes, and recovery guidance. Without state benefit-system data, the prototype must not claim to know an authoritative point-of-sale rejection reason.

3 D2: MISSION, IMPACT, AND STAKEHOLDERS

3.1 Mission Statement

WIC participants and caregivers can face uncertainty when a product appears eligible in an app but is rejected at checkout, a setting in which participant research has documented difficulty identifying eligible foods and stigma during shopping and checkout [1, 2]. That conflict can add confusion and friction during a household shopping trip. We propose extending WolfBite with an item-level explanation-and-recovery workflow that connects the rejected item's UPC and package details, the participant's current benefit or food category, and available eligibility or purchase evidence. For a bounded set of mocked mismatch scenarios, the workflow will distinguish app evidence from facts that only a retailer or WIC agency can confirm, then show rule-based likely causes and actionable next steps. This guidance is intended to help participants and caregivers choose a next action after an unexpected rejection while keeping uncertain information visible.

3.2 Measurable Claim and M0 Evaluation

M0 measurable claim: During Project 2 M0, we will test whether the explanation-and-recovery workflow produces a task-success rate of at least 80% and at least 20 percentage points higher than

Table 1: D1 validated WIC rivals.

Rival	Evidence it is real	What it does	Relevant limitation
WICShopper	Official App Store listing and vendor purchase-history documentation [3, 4]	Shows benefits, scans eligibility, provides food guidance, and records item/category purchase history in supported agencies	No documented workflow connects a scan/checkout mismatch to its likely cause and recovery steps.
myWIC Mosaic	Official App Store listing [6]	Shows benefits, tracks purchases, scans products, and supports shopping, appointments, and certification	No documented item-level explanation and recovery workflow for an unexpectedly uncovered item.
Bnft	Official App Store listing [5]	Shows balances, scans benefit eligibility, manages cards, and provides transaction history	No documented item-level reconciliation or explanation for conflicting scan and checkout results.

the baseline workflow. A trial succeeds when a test user selects the prepared scenario’s reference next step within 60 seconds without facilitator help. The baseline is a control version of the current WolfBite flow that shows the pre-checkout eligibility result and a mocked checkout rejection but provides no explanation or recovery guidance. The new and baseline flows will use the same counterbalanced, synthetic mismatch scenarios, so the evaluation requires neither live WIC data nor retailer point-of-sale access. These values are project targets to be measured at M0, not achieved results.

3.3 Stakeholders

The four Prompt 5 outputs used different labels for related roles. To avoid inflating the list with synonyms, Table 2 includes a stakeholder group only when at least two analysts named that role or a direct equivalent; one-model-only generic roles were excluded.

Table 2: Overlapping Prompt 5 stakeholders and requirements consequences.

Stakeholder	Need or fear	Design response
WIC participants, caregivers, and household representatives	Misleading eligibility guidance, uncertainty after a rejection, or the wrong household state	Identify the active household context and show evidence limits, likely causes, and a short recovery action for the selected item.
Infants, children, and pregnant or postpartum benefit recipients	Eligibility or nutrition guidance could be mistaken for advice about what is healthy or appropriate	Keep coverage guidance separate from health claims and represent missing information as unknown rather than zero or safe.
WIC clinic staff, program administrators, and agencies	Extra support burden or a prototype mistaken for official WIC authority	Show data source and freshness, and route unresolved cases to the appropriate agency contact.
Cashiers, authorized retailers, and store managers	Checkout delay, disputes, or incorrect UPC and package information when the app and register disagree	Treat the register outcome as authoritative, provide concise recovery guidance, and make no payment or point-of-sale integration claim.
State WIC IT teams, EBT processors, and card-system vendors	Unsupported integrations, security risks, or simulated balances presented as official data	Use mock adapters and simulated data, document the integration boundary, and make no live balance or payment claim.
Approved-product-list stewards, manufacturers, and product-data providers	Stale, copied, misattributed, or misidentified UPC, package, and category records	Retain source/version provenance and test stale, duplicate, and conflicting records before presenting a likely cause.
Privacy, security, and legal reviewers	Exposure of household, benefit, receipt, location, or diagnostic data and liability from unsupported claims	Minimize stored data, exclude card credentials, use privacy-safe logs, and state the prototype's limits at decision points.
Program regulators and compliance personnel	Misrepresented eligibility, unnecessary data collection, or exclusion of protected users	Trace eligibility statements to their source and test explicit privacy, accessibility, and nondiscrimination requirements.
Participants with disabilities or limited English proficiency, and their advocates	Camera-only controls, color-only status, small targets, or unclear language could block the recovery task	Preserve manual entry, pair color and icons with text, use semantic labels and plain language, and test text scaling and assistive navigation.
Receipt OCR and other outside data-service providers	Sensitive uploads, unsupported formats, or use outside service terms	Keep OCR outside the core recovery path; if retained, require consent, validate inputs, minimize retention, and provide a mock service.
Application developers and maintainers	Unreproducible failures, sensitive logs, or one-month scope growth	Use a bounded rule set, deterministic fixtures, and privacy-safe diagnostics.
Project sponsors, funders, and procurement decision-makers	Misleading public-benefit claims, hidden integration costs, or a prototype that cannot be maintained	Define a narrow pilot, measurable outcomes, and explicit external dependencies rather than promising statewide deployment.

4 D3: ENGINEERING PLAN

4.1 Before: Project 1a

- *[Add three to five completed, evidenced milestones.]*

4.2 Now: One-Month Build and Test

- *[Add three to five ambitious but checkable milestones.]*

4.3 Future: After the One-Month Project

- *[Add three to five credible future milestones.]*

4.4 Support Material and Constraints

5 D5: LLM ANALYSIS AND VERIFICATION

5.1 Models, Prompts, and Evidence

[Identify every model and runner, the required market-survey and red-team runs, the team's original keeper prompt, and where raw transcripts or screenshots are preserved. Explain the local-model result or its absence.]

5.2 Prompt-by-Model Comparison

5.3 Wrong Outputs and How We Caught Them

[Describe concrete errors, the evidence that exposed each one, and how the correction changed the rival list or requirements. Zero caught errors would not demonstrate checking.]

Table 3: Regulations, standards, licenses, and domain material.

Source	Type	Priority	Design area affected	Requirement or check created
[Insert cited authoritative sources.]				

Table 4: Comparison of the market-survey and red-team prompts.

Model	Prompt	Most useful result	Weakness or disagreement	Team adjudication
[Insert one evidence-backed row per model and required prompt.]				

5.4 Most and Least Useful Prompts

[Name the most and least useful prompts and explain why using actual outputs, not generic model commentary.]

5.5 Model Strengths and Weaknesses

[Give one concise, evidence-backed strengths/weaknesses line per model. Highlight useful disagreement rather than claiming universal agreement.]

6 CONCLUSION

[Restate the verified gap, product decision, measurable benefit, and why the one-month plan is both ambitious and testable.]

REFERENCES

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[7] Summer J. Weber, Elyse Shearer, Shelagh A. Mulvaney, Douglas Schmidt, Chris Thompson, Jessica Jones, Haseeb Ahmad, Martina Coe, and Pamela C. Hull. 2021. Prioritization of Features for Mobile Apps for Families in a Federal Nutrition Program for Low-Income Women, Infants, and Children: User-Centered Design Approach. *JMIR Formative Research* 5, 7 (2021), e30450. <https://doi.org/10.2196/30450>